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Problem Set III

1. Consider the following Dickey-Fuller regression:

$$\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \varepsilon_t,\ t = 1, 2, \ldots, T.$$

We have learned in class that it is of interest to test $H_0 : \delta = \beta = 0$ against $H_1 : \delta < 0$. Show that if β is not restricted, under the null it will accumulate into a quadratic trend in y_t.

If we only put our restriction on $\delta = 0$ under H_0, then we have:

$y_t = y_{t-1} + \alpha + \beta t + \epsilon_t = (y_{t-2} + \alpha + \beta(t-1) + \epsilon_{t-1}) + \alpha + \beta t + \epsilon_t$

If we continue recursively, we have:

$$y_t = y_0 + t\alpha + \beta \sum_{j=1}^{t} j + \sum_{i=1}^{t} \epsilon_i = y_0 + t\alpha + \frac{t(t+1)}{2}\beta + \sum_{i=1}^{t} \epsilon_i$$

Which shows the quadratic trend in null hypothesis.

2. Consider the $AR(p)$ model:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \varepsilon_t,$$

where $\varepsilon_t \sim IID\,(0, \sigma^2)$.

(a) Represent the model using the lag operator, L, and show that the existence of a unit root $z = 1$ means $\sum_{i=1}^{p} \phi_i = 1$.

(b) Show that the differenced series $\{\Delta y_t\}$ can be written as

$$\Delta y_t = \phi_1^* y_{t-1} + \phi_2^* \Delta y_{t-1} + \ldots + \phi_p^* \Delta y_{t-p+1} + \varepsilon_t,$$

where $\phi_1^* = \sum_{i=1}^{p} \phi_i - 1$, and $\phi_j^* = -\sum_{i=j}^{p} \phi_i$, for $j = 2, \ldots, p$. Therefore, testing whether $\{y_t\}$ contains a unit root is equivalent to testing whether $\phi_1^* = 0$.

(a)

$$\left(1 - \phi_1 L - L^2 - \cdots - \phi_p L^p\right) y_t = \epsilon_t \Rightarrow 1 - \phi_1 z - \phi_2 z^2 \ldots - \phi_p z^p = 0$$

To have a unit root the following equation should be zero for $z = 1$.

$1 - \phi_1 z - \phi_2 z^2 \ldots - \phi_p z^p = 0 \Rightarrow 1 - \phi_1 - \phi_2 - \cdots - \phi_p = 0 \Rightarrow \phi_1 + \phi_2 + \cdots + \phi_p = 1$

$$\Rightarrow \sum_{i=1}^{p} \phi_i = 1$$

(b)

$\Delta y_t = y_t - y_{t-1} \Rightarrow \Delta y_t = (\phi_1 - 1) y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t$

Now we have:

$\Delta y_{t-1} = y_{t-1} - y_{t-2} \Rightarrow y_{t-2} = y_{t-1} - \Delta y_{t-1}$

And $y_{t-3} = y_{t-2} - \Delta y_{t-2} = y_{t-1} - \Delta y_{t-1} - \Delta y_{t-2}$

Doing recursively, and substituting back into Δy_t, we have:

$$\Delta y_t = (\phi_1 - 1) y_{t-1} + \phi_2 (y_{t-1} - \Delta y_{t-1}) + \phi_3 (y_{t-1} - \Delta y_{t-1} - \Delta y_{t-2}) + \cdots + \phi_p \left(y_{t-1} - \sum_{i=1}^{p-1} \Delta y_{t-i} \right) + \epsilon_t$$

When we regroup the terms, we will have:

$$\Delta y_t = \left(\sum_{i=1}^{p} \phi_i - 1 \right) y_{t-1} - \sum_{i=1}^{p-1} \left(\sum_{j=i+1}^{p} \phi_j \right) \Delta y_{t-i} + \epsilon_t$$

$\phi_1^* = \sum_{i=1}^{p} \phi_i - 1$, and $\phi_j^* = -\sum_{i=j}^{p} \phi_i \ \ for\ j = 2,3, \ldots, p$

$\Rightarrow \Delta y_t = \phi_1^* y_{t-1} + \phi_2^* \Delta y_{t-1} + \cdots + \phi_p^* \Delta y_{t-p+1} + \epsilon_t$

From part a, we know that y_t has unit root, if $\sum_{i=0}^{p} \phi_i = 1$, which means that $\phi_1^* = 0$

3. Consider the model for y_t:

$$(1 - L)\, y_t = -1.2 + (1 - 0.6L)\, \varepsilon_t,$$

where $y_0 = 1$ and $\varepsilon_t \sim IID\,(0, 1)$.

(a) Classify the model as $ARIMA(p, d, q)$.

(b) Find the mean of y_t, for $t \geq 0$. Does it converge to a finite limit for t large?

(c) Consider another model: $(1 - 0.95L)\, x_t = -1.2 + (1 - 0.6L)\, \varepsilon_t$, where $x_0 = 1$ and $\varepsilon_t \sim IID\,(0, 1)$. Explain why the mean of x_t converges to a limit for t large and find the limit.

(a)

ARIMA(0,1,1)

(b)

$\mathbb{E}[y_t] = -1.2 + \mathbb{E}[y_{t-1}]$

$\mathbb{E}[y_{t-1}] = -1.2 + \mathbb{E}[y_{t-2}]$

Continuing recursively, we have:

$\mathbb{E}[y_t] = y_0 - 1.2t = 1 - 1.2t$

Which is not converging to a finite limit for large t

(c)

This process is stationary modeled by ARIMA(1,0,1), which its AR part has roots greater than 1 or $\frac{1}{0.95}$, and MA part is finite, so it is stationary. Therefore it has a finite mean. And the mean is:

$$-\frac{1.2}{1-0.95} = -24$$

4. Consider the model:

$$y_t = (1+0.5L)\left(1+0.3L^4\right)\varepsilon_t, \quad \varepsilon_t \sim WN\left(0,\sigma^2\right).$$

(a) Is $\{y_t\}$ weakly stationary? Is it invertible?

(b) Find the ACF of $\{y_t\}$.

(c) Compare your finding in part (b) with the ACFs of models $y_t = (1+0.5L)\varepsilon_t$ and $y_t = \left(1+0.3L^4\right)\varepsilon_t$.

(a)

$y_t = \epsilon_t + 0.5\epsilon_{t-1} + 0.3\epsilon_{t-4} + 0.15\epsilon_{t-5}$

Since it is a finite MA process, it is weakly stationary. Also, since the roots of MA process all lie outside of unit circle, it is invertible.

(b)

$$\gamma_0 = (1 + 0.25 + 0.09 + 0.0225)\sigma^2 \Rightarrow \rho_0 = \frac{\gamma_0}{\gamma_0} = 1$$

$$\gamma_1 = \mathbb{E}[y_t y_{t-1}] = (0.5 + 0.045)\sigma^2 \Rightarrow \rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{(0.5 + 0.045)}{(1 + 0.25 + 0.09 + 0.0225)} = 0.4$$

$$\gamma_2 = 0 \Rightarrow \rho_2 = \frac{\gamma_2}{\gamma_0} = 0$$

$$\gamma_3 = \mathbb{E}[y_t y_{t-3}] = 0.15\sigma^2 \Rightarrow \rho_3 = \frac{\gamma_3}{\gamma_0} = \frac{0.15}{(1 + 0.25 + 0.09 + 0.0225)} \approx 0.11$$

$$\gamma_4 = \mathbb{E}[y_t y_{t-4}] = (0.3 + 0.075)\sigma^2 \Rightarrow \rho_4 = \frac{\gamma_4}{\gamma_0} = \frac{(0.3 + 0.075)}{(1 + 0.25 + 0.09 + 0.0225)} \approx 0.275$$

$$\gamma_5 = \mathbb{E}[y_t y_{t-5}] = 0.15\sigma^2 \Rightarrow \rho_5 = \frac{\gamma_5}{\gamma_0} = \frac{0.15}{(1 + 0.25 + 0.09 + 0.0225)} \approx 0.11$$

$$\gamma_h = 0, h > 5 \Rightarrow \rho_h = 0, h > 5$$

(c)

For $y_t = (1 + 0.5L)\epsilon_t$, ACFs are:

$$\gamma_0 = (1 + 0.25)\sigma^2 \Rightarrow \rho_0 = \frac{\gamma_0}{\gamma_0} = 1$$

$$\gamma_1 = 0.5\sigma^2 \Rightarrow \rho_1 = \frac{0.5}{1.25} = 0.4$$

$$\gamma_h = 0, h \geq 2 \Rightarrow \rho_h = \frac{\gamma_h}{\gamma_0} = 0 \; for \; h \geq 0$$

For $y_t = (1 + 0.3L^4)\epsilon_t$, ACFs are:

$$\gamma_0 = (1 + 0.09)\sigma^2 \Rightarrow \rho_0 = \frac{\gamma_0}{\gamma_0} = 1$$

$$\gamma_{1,2,3} = 0 \Rightarrow \rho_{1,2,3} = 0$$

$$\gamma_4 = 0.3\sigma^2 \Rightarrow \rho_4 = \frac{0.3}{1.09} \approx 0.275$$

$$\gamma_h = 0, h \geq 5 \Rightarrow \rho_h = \frac{\gamma_h}{\gamma_0} = 0 \; for \; h \geq 5$$

As we expected in part b, we have all ACF of part c plus some more as we have some cross multiplications.